Butcher

Game and source code

The game mechanics are completely implemented, very good.

WaitVBlank: use the z flag, not c.

LoadTitle and LoadEnd should be a single function that takes hk and b as parameters.

Make the MainLoop shorter by defining and calling functions, or moving code into functions, for instance:

```
.movement:
  ld a, [isFalling]
  or a
  jr nz, .do_gravity
  call MovePlayer
.do_gravity:
```

The “isFalling” check can happen inside of MovePlayer

Report

In this course we just only about instructions cycles. No such thing as T-states and M-cycles. So please stick to the vocabulary we use in the course.

Current grade: 17 (including -1 due to Assignment 2)

CatGirlHunter

Game and source code

The game implement all the required mechanics and the source code is very good.

Functions MaybeNextLevel and MaybePreviousLevel could be merged into a single MaybeChangeLevel function that checks for A and B key press, that would make the MainLoop a bit shorter.

You can define LEVEL_COUNT as the actual number of levels in the ROM instead of hardcoding it as 10.

Report

I already talked about the AI-generated nonsense during the presentation. Please fix the report.

In this course we do not use words like “dots” or “m-cycles”, we just talk about cycles.

The Runtime Data Flow chart is good.

Current Grade: 15

Catrap123

Game and source code

The game mechanics are correctly implemented.

Remove LevelTable and take advantage of the fact that all levels have the same size (see maze.asm on Moodle).

Since you are using only one object, it is not necessary to use ShadowOAM, you can use variables for the player's coordinates and that is all. The current code uses more cycles than allowed during VBlank.

In MainLoop, move the code between "call WaitVBlank" and " call CopyShadowOAMtoOAM " into its own function to improve organization (also

wKeyHeld is not used and wKeyPressed is already the current variable.

Report

The report should explain what tool you used to create the tiles:

```
  db $00,$00,$00,$00,$00,$00,$00,$00  ; Tile 0: NOTHING
  db $00,$FF,$00,$FF,$00,$FF,$00,$FF  ; Tile 1: STAIRS
  db $3C,$42,$5A,$42,$42,$5A,$42,$3C  ; Tile 2: MONSTER
  db $3C,$42,$81,$81,$81,$81,$42,$3C  ; Tile 3: ROCK
  db $AA,$55,$AA,$55,$AA,$55,$AA,$55  ; Tile 4: SAND
  db $FF,$FF,$FF,$FF,$FF,$FF,$FF,$FF  ; Tile 5: WALL
  db $3C,$7E,$DB,$FF,$FF,$FF,$A5,$A5  ; Tile 6: GHOST
  db $81,$42,$3C,$5A,$7E,$42,$42,$3C  ; Tile 7: PLAYER
```

Update the cycles calculation after ShadowOAM and CopyShadowOAMtoOAM.

Also, the variables "previous,current" have nothing to do with key debouncing.

Current grade: 16

FlashSync

Game and source code

The game mechanics are correctly implemented.

Missing features: Start to reset the current level and A/B to change levels.

Since you only use the shadowOAM for the player's coordinates, it's better to just define variables for the player's X and Y coordinates, and not use shadowOAM at all.

In the following snippet:

```
    jp ScreenSwitching ; 16
    ret ; 16  <-  this ret is dead code, remove it
```

CopyTilesToVRAM can be optimized by using post-increment instructions (ld [hl+],a).

Report

You should explain what tools you used to generate these tiles that are included in hexadecimal (0 to 7).

Current grade: 15

KingBob

Game and source code

The game implements the original mechanics correctly.

The hold-and-repeat input is a nice touch.

Please make the variables declarations less expansive vertically (one line per variable is enough).

Replace magic values such as $9800 by constants from hardware.inc. Also addresses like $98C7 can be written as TILEMAP0+ (some number) for more clarity (this informs the reader that the address belongs to the background).

If there is no concrete reason to do otherwise, just use the WaitVBlank function we always used in the Lab instead of the current version you have.

You can define NUM_LEVELS as the actual number of levels in the ROM instead of hardcoding it as 10 (see maze.asm on Moodle).

You can do away with TileBufferRowTable and BGRowTable with some arithmetic.

Report

Explain what tools you used to create the tiles written as hexadecimal in the source code.

Do not mention in the report things like “No pre-existing game engine or framework is used. The program does not use DMA, interrupt handlers, or rst instructions“, it’s pointless.

The report should justify the value of DIRTY_MAX. In the source code the value is 32, in the report it says 4 is enough, so what should it be? Update the cycles count accordingly.

Current grade: 8

Lebang

The game mechanics are correctly implemented.

The source code is short and well organized, and the data structure are simple. Good choices have been made.

A detail, this:
```
  ld   d, 0
  ld   e, 32 - LEVEL_W
```
can be turned into: ld de, 32 - LEVEL_W.

Report

In the “Keys” table, remove the column “Key on computer” since it depends on the emulator.

In “Data Structure”, no need to mention OAM, it is part of the hardware, not of your program.

The report should mention that the effect of gravity applies across several frames, since gravity is processed from top to bottom, this has the side effect of creating a (fast) fall animation.

Current grade: 18

WWWWW

Game and source code

The game implements the original game mechanics correctly.

Reset should reset only the current level, not take you back to the title screen.

You can merge MonNum and GhoNum into a single variable.

The Level1 to Level10 functions can be replaced by a standard function that loads any level and takes the level number as parameter. All levels have the same size. Get rid of all the CopyLevelNToLogicMap and merge it into a single function. Your source code size will be greatly reduced.

You don't need to use ShadowOAM just for one object, just use the player's X and Y coordinates.

Reorganize the MainLoop. Instead of calling the (generically called) Update function, move the contents of Update into the MainLoop, it makes the loop easier to follow.

DrawLogicMap can be made faster, instead of:

```
   push de
   ld de, 20
   add hl, de
   pop de
```

Do:

```
        ld a,b
        ld bc, 20
        add hl,bc
        ld b,a
```

Furthermore you may set LogicMap at the beginning of a memory page so instead of "inc de" you can do "inc e" which saves 4 cycles per copied byte.

Report

An inconsistency: it's not clear why the report mentions that "CopyShadowOAMtoOAM copies 8 OAM entries" when it also says that only one is used. Please fix that.

Current grade: 13